Determinants

Lecture

Determinants of second order

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}\, a_{22} - a_{21}\, a_{12}$$

Example

Evaluate the following determinants:

(i) $\begin{vmatrix} 3 & 4 \\ 6 & 5 \end{vmatrix}$ (ii) $\begin{vmatrix} 2 & -2 \\ 1 & 4 \end{vmatrix}$

Solution

(i) $\begin{vmatrix} 3 & 4 \\ 6 & 5 \end{vmatrix} = 15 - 24 = -9$

(ii) $\begin{vmatrix} 2 & -2 \\ 1 & 4 \end{vmatrix} = 8 - (-2) = 8 + 2 = 10$

Determinants of third order

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

One way in which it can be evaluated is to use second-order determinants as follows:

$$a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

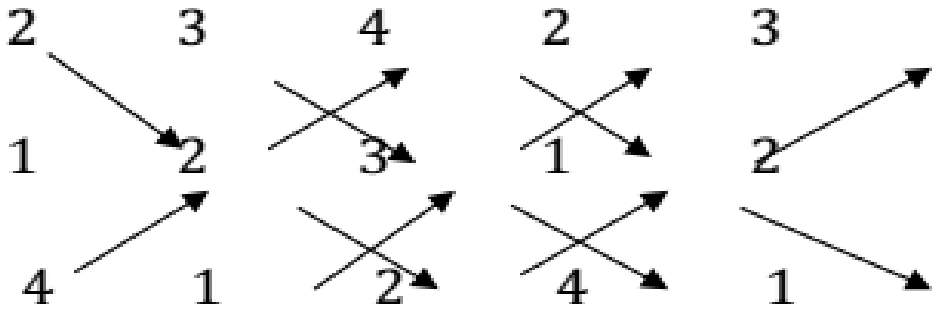
Example

$$\begin{vmatrix} 2 & 3 & 4 \\ 1 & 2 & 3 \\ 4 & 1 & 2 \end{vmatrix} = 2\begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} - 3\begin{vmatrix} 1 & 3 \\ 4 & 2 \end{vmatrix} + 4\begin{vmatrix} 1 & 2 \\ 4 & 1 \end{vmatrix}$$

$$= 2\,(4-3) - 3\,(2-12) + 4\,(1-8)$$

$$= 2 + 30 - 28 = 4$$

Another solution Using sarrus method



$$\Delta = (2 * 2 * 2 + 3 * 3 * 4 + 4 * 1 * 1)$$

$$-(4 * 2 * 4 + 1 * 3 * 2 + 2 * 1 * 3)$$

$$\Delta = (8 + 36 + 4) - (32 + 6 + 6)$$

$$= 48 - 44 = 4$$

Cramer's Rule

Cramer's Rule- two equations

If we are given a pair of simultaneous equations

$$a_{11}\,x + a_{12}\,y = b_1$$

$$a_{21}\,x + a_{22}\,y = b_2$$

Then x, and y can be found from

$$x = \frac{\Delta_x}{\Delta}, \qquad y = \frac{\Delta_y}{\Delta}$$

where,

$$\Delta = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}, \quad \Delta_x = \begin{vmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{vmatrix}, \quad \Delta_y = \begin{vmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{vmatrix}$$

Example

Solve the equations

$$3x + 4y = -14, \quad -2x - 3y = 11$$

Solution

$$\Delta = \begin{vmatrix} 3 & 4 \\ -2 & -3 \end{vmatrix} = -1, \quad \Delta_x = \begin{vmatrix} -14 & 4 \\ 11 & -3 \end{vmatrix} = -2,$$

$$\Delta_y = \begin{vmatrix} 3 & -14 \\ -2 & 11 \end{vmatrix} = 5$$

$$\therefore x = \frac{\Delta_x}{\Delta} = \frac{-2}{-1} = 2, \qquad y = \frac{\Delta_y}{\Delta} = \frac{5}{-1} = -5$$

2. Cramer's Rule - three equations

For the case of three equations in three unknowns: If

$$a_{11}\,x + a_{12}\,y + a_{13}\,z = b_1$$

$$a_{21}\,x + a_{22}\,y + a_{23}\,z = b_2$$

$$a_{31}\,x + a_{32}\,y + a_{33}\,z = b_3$$

Then x, y and z can be found from

$x = \frac{\Delta_x}{\Delta}$, $y = \frac{\Delta_y}{\Delta}$, $z = \frac{\Delta_z}{\Delta}$, where

where

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, \quad \Delta_x = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}$$

$$\Delta_y = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, \quad \Delta_z = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$$

Example

Solve the system

$$4x + 2y - z = 5,\ 3x - y - z = -2, \qquad x + y - z = 0$$

Solution

$$\Delta = \begin{vmatrix} 4 & 2 & -1 \\ 3 & -1 & -1 \\ 1 & 1 & -1 \end{vmatrix} = 8 \neq 0\ ,\ \Delta_x = \begin{vmatrix} 5 & 2 & -1 \\ -2 & -1 & -1 \\ 0 & 1 & -1 \end{vmatrix} = 8$$

$$\Delta_y = \begin{vmatrix} 4 & 5 & -1 \\ 3 & -2 & -1 \\ 1 & 0 & -1 \end{vmatrix} = 16\ ,\ \Delta_z = \begin{vmatrix} 4 & 2 & 5 \\ 3 & -1 & -2 \\ 1 & 1 & 0 \end{vmatrix} = 24$$

$$\therefore x = \frac{\Delta_x}{\Delta} = \frac{8}{8} = 1\ , y = \frac{\Delta_y}{\Delta} = \frac{16}{8} = 2, \qquad z = \frac{\Delta_z}{\Delta} = \frac{24}{8} = 3$$

Therefore, $x = 1, y = 2$ and $z = 3$.

- **Properties of Determinants**

Property 1: If one row of a matrix consists entirely of zeros, then the determinant is zero.

Ex: $\begin{vmatrix} 0 & 0 & 0 \\ 3 & 2 & 1 \\ 1 & 3 & 5 \end{vmatrix} = 0$

Property 2: If two rows of a matrix are interchanged, the determinant changes sign.

Ex: $\begin{vmatrix} 4 & 2 & -1 \\ 3 & -1 & -1 \\ 1 & 1 & -1 \end{vmatrix} = -\begin{vmatrix} 3 & -1 & -1 \\ 4 & 2 & -1 \\ 1 & 1 & -1 \end{vmatrix} = -8$

Property 3: If two rows of a matrix are identical, the determinant is zero.

Ex: $\begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 4 & 7 & 5 \end{vmatrix} = 0$

Property 6: If a matrix B is obtained from a matrix A by adding to one row of A, a scalar times another row of A, then

$$|B| = |A|.$$

Ex: $|A| = \begin{vmatrix} 4 & 2 & -1 \\ 3 & -1 & -1 \\ 1 & 1 & -1 \end{vmatrix} = 8$, [Add $R_1 + R_2$]

$$|B| = \begin{vmatrix} 4+3 & 2-1 & -1-1 \\ 3 & -1 & -1 \\ 1 & 1 & -1 \end{vmatrix}$$

$$= \begin{vmatrix} 7 & 1 & -2 \\ 3 & -1 & -1 \\ 1 & 1 & -1 \end{vmatrix} = 8$$

<u>Property 7:</u> $\det(\mathrm{A}) = \det(\mathrm{A}^T)$.

Property 8: The determinant of a triangular matrix, either upper or lower, is the product of the elements on the main diagonal.

Ex: $\begin{vmatrix} 1 & -4 & 3 \\ 0 & 2 & 5 \\ 0 & 0 & 5 \end{vmatrix} = 10, \begin{vmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & 7 & 5 \end{vmatrix} = 15$

<u>Property 9:</u> If A and B are of the same order, then $\det(\mathrm{A}\,\mathrm{B}) = \det(A)\det(B)$

Example: Prove that

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$$

Solution

Subtracting $C_2 - C_1$, $C_3 - C_1$ we get

$$\begin{vmatrix} 1 & 0 & 0 \\ a & b-a & c-a \\ a^2 & b^2-a^2 & c^2-a^2 \end{vmatrix}$$

$$= (b-a)(c-a)\begin{vmatrix} 1 & 0 & 0 \\ a & 1 & 1 \\ a^2 & b+a & c+a \end{vmatrix}$$

Subtracting $C_3 - C_2$ we get

$$= (b-a)(c-a)\begin{vmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ a^2 & b+a & c-b \end{vmatrix} = (b-a)(c-a)(c-b)$$

Example:

Solve $\begin{vmatrix} x-3 & 1 & -1 \\ 1 & x-5 & 1 \\ -1 & 1 & x-3 \end{vmatrix} = 0$

Solution

Collecting $R_1 + (R_2 + R_3)$

$$\therefore \begin{vmatrix} x-3 & x-3 & x-3 \\ 1 & x-5 & 1 \\ -1 & 1 & x-3 \end{vmatrix} = 0$$

$$\therefore (x-3)\begin{vmatrix} 1 & 1 & 1 \\ 1 & x-5 & 1 \\ -1 & 1 & x-3 \end{vmatrix} = 0$$

Subtracting $C_2 - C_1$, $C_3 - C_1$ we get

$$\therefore (x-3)\begin{vmatrix} 1 & 0 & 0 \\ 1 & x-6 & 0 \\ -1 & 2 & x-2 \end{vmatrix} = 0$$

$$\therefore (x-3)(x-6)(x-2)$$

$$\therefore x = 2, 3, 6$$

Proved that $\begin{vmatrix} \ln x & \ln y & \ln z \\ \ln\ 2x & \ln\ 2y & \ln\ 2z \\ \ln\ 3x & \ln\ 3y & \ln\ 3z \end{vmatrix} = 0$

Solution

Subtracting $R_2 - R_1$, $R_3 - R_1$ we get

$$.H.S = \begin{vmatrix} \ln x & \ln y & \ln z \\ \ln\ 2x - \ln x & \ln\ 2y - \ln y & \ln\ 2z - \ln z \\ \ln\ 3x - \ln x & \ln\ 3y - \ln y & \ln\ 3z - \ln z \end{vmatrix}$$

$$= \begin{vmatrix} \ln x & \ln y & \ln z \\ \ln\ 2 & \ln\ 2 & \ln\ 2 \\ \ln\ 3 & \ln\ 3 & \ln\ 3 \end{vmatrix}$$

$$= \ln\ 2 . \ln 3 \begin{vmatrix} \ln x & \ln y & \ln z \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= \ln\ 2 . \ln 3\,(0)$$

$$= 0 = R.H.S$$

Example : Prove that

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (b-c)(c-a)(a-b)(a+b+c)$$

Solution

Subtracting $C_1 - C_2$, $C_2 - C_3$ we get

$$L.H.S = \begin{vmatrix} 0 & 0 & 1 \\ a-b & b-a & c \\ a^3-b^3 & b^3-c^3 & c^3 \end{vmatrix}$$

$$= (a-b)(b-c)\begin{vmatrix} 0 & 0 & 1 \\ 1 & 1 & c \\ a^2+ab+b^2 & b^2+bc+c^2 & c^3 \end{vmatrix}$$

$$= (a-b)(b-c)\begin{vmatrix} 1 & 1 \\ a^2+ab+b^2 & b^2+bc+c^2 \end{vmatrix}$$

$$= (a-b)(b-c)\,[b^2+bc+c^2-\ a^2-ab-b^2\,]$$

$$= (a-b)(b-c)\,[b\,(c-a)+(c-a)\,(c+a)]$$

$$= (a-b)(b-c)(c-a)(b+c+a)$$

$$= (b-c)(c-a)(a-b)(a+b+c)$$

$$= R.H.S$$

Example : Prove that

$$\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = 2(a+b+c)^3$$

Solution

Adding $C_1 + (C_2 + C_3)$

$$L.H.S = \begin{vmatrix} 2(a+b+c) & a & b \\ 2(a+b+c) & b+c+2a & b \\ 2(a+b+c) & a & c+a+2b \end{vmatrix}$$

$$= 2(a+b+c)\begin{vmatrix} 1 & a & b \\ 1 & b+c+2a & b \\ 1 & a & c+a+2b \end{vmatrix}$$

Subtracting

$R_2 - R_1$

$R_3 - R_1$

$$2(a+b+c)\begin{vmatrix} 1 & a & b \\ 0 & a+b+c & 0 \\ 0 & 0 & a+b+c \end{vmatrix}$$

$$= 2(a+b+c)(a+b+c)(a+b+c)$$

$$= 2(a+b+c)^3$$

$$= R.H.S$$

thank you